Look at the equation again → heat transfer coefficient

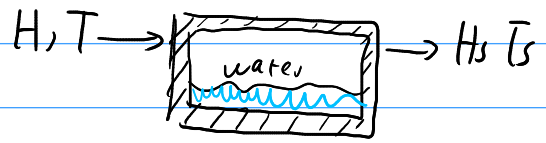
$$\frac{H - H_w}{T - T_w} = -\frac{h/(M_B k_y)}{\lambda_w}$$

↑ w: wet-bulb

Ratio $\frac{h}{M_B k_y}$ = psychrometric ratio (psychro- = cold, metric = measure) measure wet bulb

In the adiabatic (no external heat exchange) process

Heat balance:



saturated temperature T_s

Coincidentally, $\frac{h}{M_B k_y} \simeq 0.96 - 1.025$

$C_s \simeq 1.005 + 1.88 \cdot H$

Use the "adiabatic line" ↓ T_w

Let's use the humidity chart for some analysis

Humid air at 40°C has a wet bulb temp of 20°C

what is H in current air, and calulate H_p, Dew point, V_H humid heat & enthalpy

$H(20°C, sat) = 0.0147$ kg w/kg air

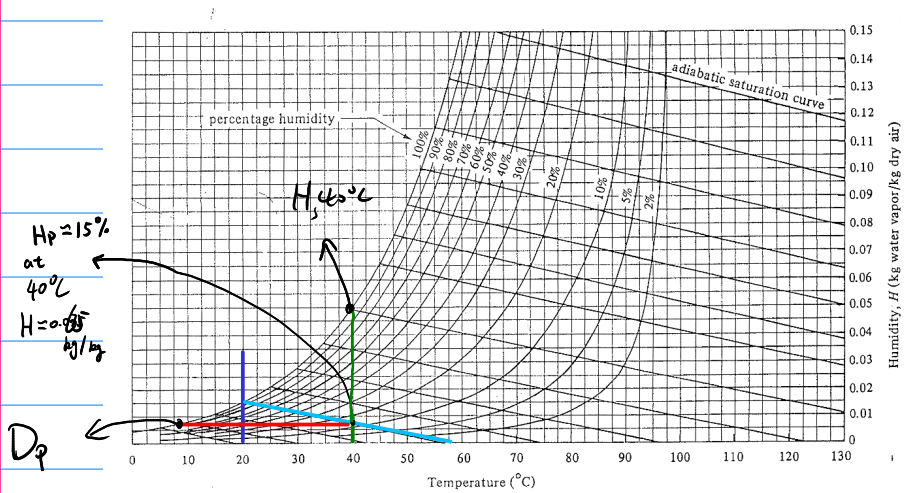


FIGURE 9.3-2. *Humidity chart for mixtures of air and water vapor at a total pressure of 101.325 kPa (760 mm Hg). (From R. E. Treybal, Mass-Transfer Operations, 3rd ed. New York: McGraw-Hill Book Company, 1980. With permission.)*

Follow the chart

$$\frac{H - H_w}{T - T_w} = -\frac{1.005 + 1.88H}{\lambda_s}$$

or to solve

$$C_s(T - T_s) + H\lambda_s = C_s(T_s - T_s) + H_s\lambda_s$$

$H(40°C) = 0.0065$

$$\frac{H - H_w}{T - T_w} = -\frac{1.005 + 1.88H}{2501.4}$$

Percentage humidity

at 40°C $\quad H(40°C) = 0.049$

$$H_p = \frac{0.0065}{0.049} = 13\%$$

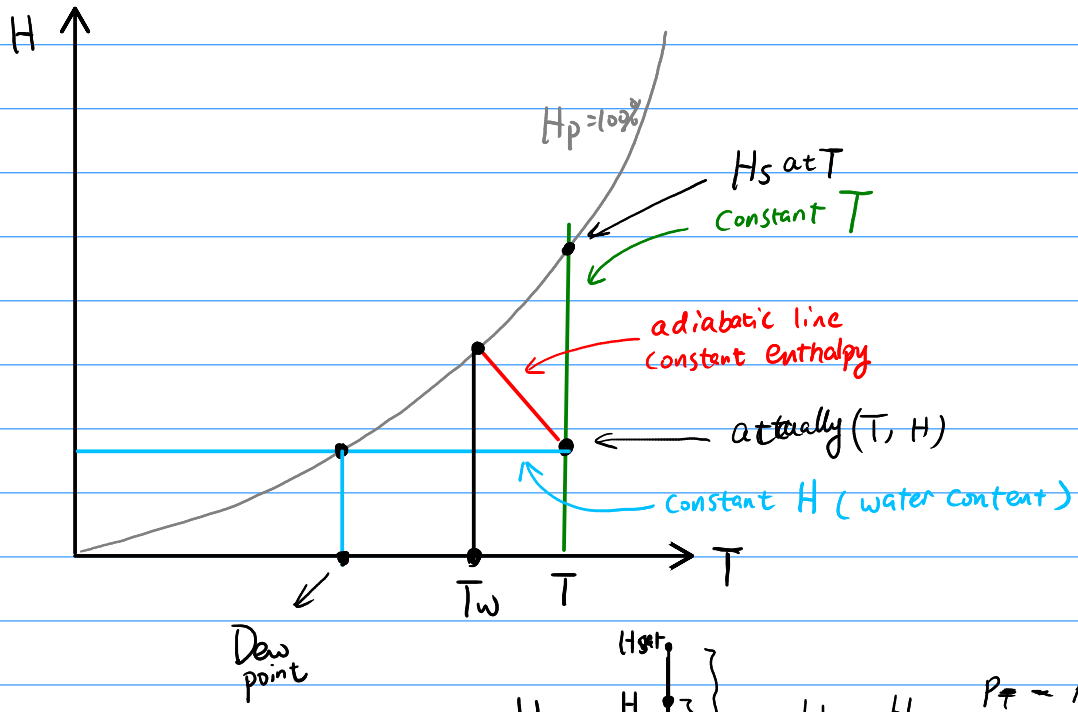
Relative humidity

$$H_R = \frac{P_A}{P_T - P_A} \simeq 14\%$$

Dew point : around 8°C

$V_H \doteq 0.90\ m^3/kg$ dry air $\quad$ (use H at 40°C)

$C_s = 1.005 + 1.88H \doteq 1.02\ kJ/(\text{dry air})/$

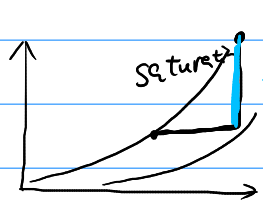


H_p: H_{sat}, H, 0 $\quad H_p = H_R \cdot \dfrac{P_T - P_{vap}}{P_T - P_A}$

H_p slight smaller than H_R

What is the indication of the wet bulb temperature ?

1. cooling of water using dry air is possible (max cooling T_w)
2. vapour pressure difference is driving force (see previous discussion of humid hot)



→ driving force

this is the concept behind cooling tower

CHE318 L29

Cooling tower introduction

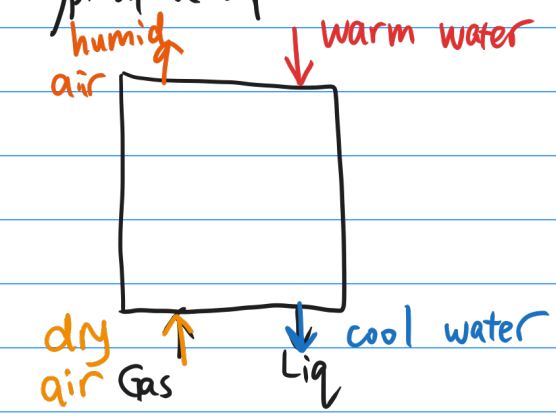
Cooling tower takes the principle of humidification process

Put more water into air ⟶ requires heat

↑

absorbes heat from liquid

Typical setup



why cooling?

① dry air → evaporation (below H_{vap})

② latent heat flow: from water → air

① Minimal temp during cooling? T_w

② Why evaporation occurs?

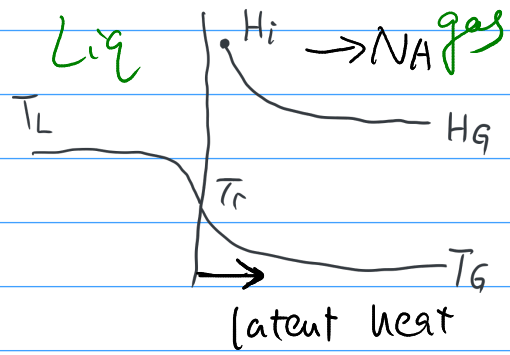
$$P_{vap}|_T - P_A|_{T_w} > 0$$

③ usual setup: get water flow rate & constant air flow can be forced (like blowing a cup of hot water)

④ sensitive to air-water contact

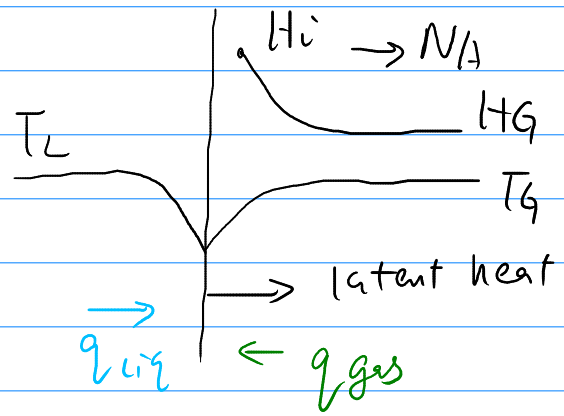
Design analysis for cooling tower

Because both heat / mass transfer occur, draw diagram

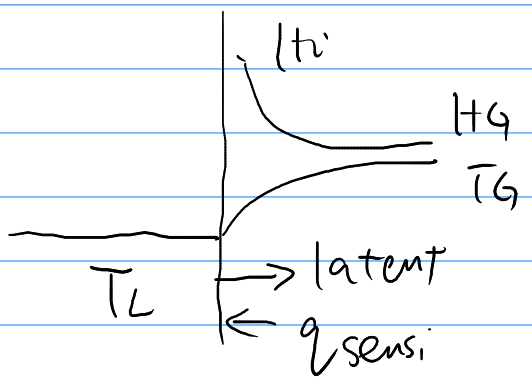


Top of column

(T_G is low)



middle of column



mass transfer always liq → gas

but T profile different

How to solve?

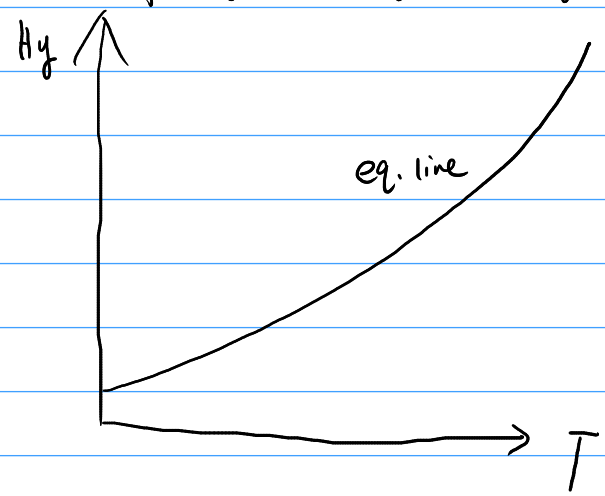
want to know

① Given liq flow rate, gas flow rate ⇒ final T

③ Design tower height

We have similar operating line in cooling towers

Instead of using humidity, use H_y (enthalpy of mixture)



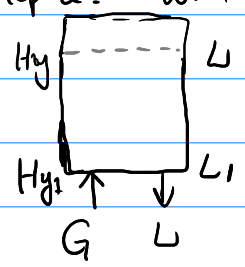
← $H_y = (1.005 + 1.88 H_s) + H_s \cdot 2501.4$ kJ/kg

H_s at each T ← from psychrometric chart

Careful unit!

If currently not saturated $H < H_s$, point below eq. line

Step 2: what is operating line?



Control volume from bottom to location 8

(H_{y1}, G, L_1) (H_y, L)

Since only very small amount is evaporated

$L_1 \simeq L = \text{const}$

Heat balance

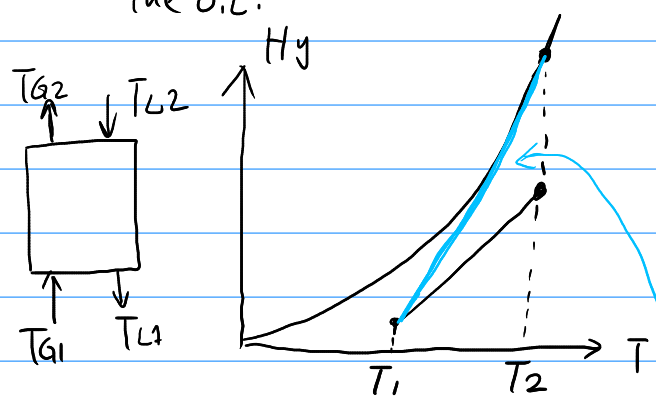
Heat in = Heat out

$$G(\underbrace{H_y - H_{y1}}_{\text{Diff in Enthalpy}}) = L \underbrace{c_L}_{\text{heat cap. in liquid}} (T_L - T_{L1})$$

$c_L = 4.187$ kJ/kg·K

$\doteq 1$ kcal/kg·K

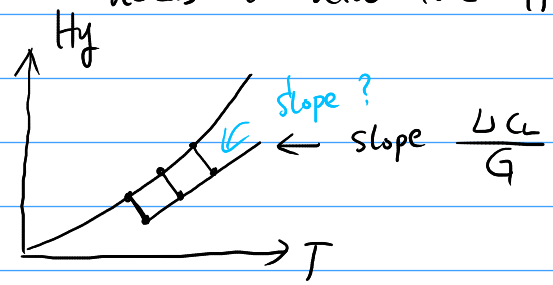
describes the O.L.



Slope? $\boxed{\dfrac{H_y - H_{y1}}{T_L - T_{L1}} = \dfrac{L c_L}{G}}$

If L G known, can calculate T_{L2} in gas

maximum slope (max L flow allowed)

analogous to min L' in 2-phase

What else do we need? Calculate height Z that needs to have the H_{y_i}, T_i at interface